

Roy Yao

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EDUCATION:

University of California, Berkeley

(Aug 2023- May 2027)

B.S. Electrical Engineering & Computer Science, Bioengineering, GPA: 3.84

HONORS/AWARDS: Dean's List, EECS Honors Program, Rose Hills Summer Scholarship, GLOBE Discovery Scholarship

Relevant Coursework: **CS 189:** Machine Learning; **CS 180:** Computer Vision; **CS 162:** Operating Systems and System Programming; **EE 120:** Signals and Systems; **EECS 106A:** Robotics; **EECS 127:** Optimization Models in Engineering; **CS 170:** Efficient Algorithms and Intractable Problems; **EE 290:** MRI, Brain Imaging; **EE 145B:** Medical Imaging Signals and Systems; **EE 198:** Micromouse; **BioE 101:** Instrumentation in Biology and Medicine; **BioE 121, BioE 121L:** BioMEMS, Medical Devices, and BioNanotechnology Laboratory

EXPERIENCES:

Imaging Scientist Intern, Varex Imaging, San Jose

(Jun 2026 - Aug 2026)

- Co-authored SPIE Medical Imaging 2027 submission on a novel charge-sharing correction scheme for photon-counting detectors; designed Geant4 Monte Carlo simulations and multi-material phantoms to validate detector response against measured data.
- Debugged charge-sharing correction logic, built a custom ImageJ analysis plugin, and developed calibration/documentation workflows adopted by the engineering team, improving simulation accuracy and reproducibility.

Undergraduate Researcher, Conolly Lab, UC Berkeley

(Aug 2025 -)

- Verified, designed, and optimized a low-noise analog front-end circuit for Magnetic Particle Imaging (MPI), including LTspice simulation and PCB implementation of a low-noise preamplifier, achieving a 10-times increase in signal-to-noise ratio (SNR) with 5 MHz bandwidth.

Summer Research Intern, Chen Lab, National Tsing Hua University

(Jun 2025 - Jan 2026)

- Participated in the 2025 Summer Research Program. Developed signal processing and correlation algorithms (bandpass filtering, lag alignment, amplitude thresholding) to validate nanoparticle transit events against lock-in amplifier outputs. Built OpenCV pipelines (pattern recognition, model training) and enabled video-based particle counting independent of photodiode DAQ systems.

Software Team Member, Neurotech@Berkeley, UC Berkeley

(Jan 2025 -)

- Utilized a CNN model and data augmentation to achieve 20x accuracy using speckle data. Applied Latent Consistency Model (LCM) to reduce runtime and enhance the clarity and reduce the runtime of reconstructed fMRI images.

Undergraduate Researcher, Ganguly Lab, UC San Francisco

(Oct 2023 - May 2024)

- Coded for empirical mode decomposition (EMD) and channel dropout data augmentation on BCI data. Wrote and applied neural network models such as multilayer perceptron (MLP), convolutional neural network (CNN), recurrent neural network (RNN), and long short-term memory (LSTM) to increase accuracy on BCI action classification.
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PROJECTS:

Robotics Manipulation & Mobile Navigation, UC Berkeley

(Aug 2026 -)

- **UR7e Robotic Arm:** Developed a ROS 2 manipulation pipeline using an Intel RealSense depth camera for 3D point-cloud perception, MoveIt inverse kinematics, and trajectory planning to position and manipulate objects with a 6-DOF robotic arm.
- **TurtleBot 3 Autonomous Navigation:** Implemented LiDAR-based mapping and probabilistic occupancy-grid updates in ROS 2, enabling the mobile robot to construct an environment representation for autonomous navigation and motion planning.

Pintos Operating System, UC Berkeley

(Jan 2026 - May 2026)

- Implementing core OS functionality including user-process management, system calls, thread synchronization, priority scheduling, and file-system operations. Built and debugged low-level kernel components for process creation, argument passing, file I/O, and concurrency, coordinating threads with locks, semaphores, and scheduling policies. Developed and tested systems code using GDB, tracing kernel state, synchronization bugs, system-call failures, and resource-management issues.

Microfluidic Single-Cell Dispensing Device, UC Berkeley

(Apr 2026 - May 2026)

- Designed and fabricated a passive microfluidic trapping device with a 9 mm-pitch layout aligned to standard 96-well plates, optimizing channel geometry, trap placement, corner radii, and hydrodynamic resistance for deterministic single-particle capture, achieving up to 83% single-bead trapping efficiency through optimization of the least-flow-resistance path and transient pressure-based trap reset.

EMG-Based Word Detection Sensor, UC Berkeley

(Apr 2025 - May 2025)

- Designed and prototyped a facial EMG sensor using Ag/AgCl electrodes, AD620 instrumentation amplifiers, and Arduino MKR Zero ADC for real-time speech-related signal acquisition.
- Implemented two-stage band-pass filtering (20–500 Hz, gain ≈ 4000) and autocorrelation-based classification augmented with cosine similarity, MSE, and CNN algorithms, achieving 80% accuracy.

Silent Speech Decoding, UC Berkeley

(Aug 2025 - Dec 2025)

- Developed a multimodal ML pipeline for silent-speech recognition using the RVTALL dataset, extracting features from audio, lip landmarks, RGB/depth video, and laser speckle while evaluating cross-modal synchronization. Trained a 1D CNN on augmented laser-speckle time-series data, improving 30-class classification of vowels, words, and sentences to 71.15% test accuracy with 0.71 F1
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SKILLS:

- **Programming:** Python, MATLAB, C++, C, Java, SQL
- **Libraries:** NumPy, PyTorch, OpenCV
- **Tools:** Git, LTspice, KiCad, COMSOL, SolidWorks, Arduino
- **Technical Areas:** Signal Processing, Machine Learning, Computer Vision, Circuit Design, Optimization